Rajiv Choudhury

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Summary

Incoming CS Graduate student at **UC Davis** (Fall 2026) and Computer Science undergraduate from IIIT Guwahati with experience in on-device AI, embedded systems, and energy-efficient computing. Interested in the intersection of **parallel computing** and **IoT/embedded devices** to enhance **ML inference efficiency** on constrained hardware such as smart watches, TVs, and digital appliances. Experienced in developing optimized AI systems and conducting research with **IISc Bengaluru** and **IBM Research**. In addition have been performing research on embedded AI for Android platform with **Samsung**.

Education

Class of 2024	B.Tech (Computer Science Engineering) at **IIIT Guwahati**	(CPI: 9.13/10)
	Relevant Coursework: Operating Systems, Computer Organisation, Machine Learning, Deep Learning, Computer Networks, Theory of Computaion, Data Structure & Algorithm	
Class of 2020	Class 12th MBS Public School (Affilitated with CBSE)	(Grade: 90.80%)
Class of 2018	Class 10th DAV Public School (Affilitated with CBSE)	(Grade: 90.60%)

Work Experience

Software Engineer — Samsung R&D Institute India - Delhi Jul 2024 – Present

- Developing and maintaining **Tizen OS** application features using **C++** and **C#**.
- Developed a **USB preset** feature for **E-paper signage devices**, enabling zero-contact device setup through configuration files.
- Implemented an early-warning feature for certificate expiration across **10,000+ enterprise displays**, which received the team's **Best Feature Award**.
- Maintained and developed multiple modules for **Tizen OS**, including the **Power Saving Module**, achieving up to **80% power savings** across enterprise displays.
- Developed and maintained a **logger service** for collecting logs from Android devices, enabling efficient post-sale diagnostics and reducing customer support effort.
- Developed an on-device AI feature integrating an offline **Speech-to-Text model (OpenAI Whisper)** on Android Interactive Displays for Samsung's AI Assistant project.
- Built Android applications for Interactive Displays using **Jetpack Compose**, improving user experience and enhancing application performance by **25%**.
- Developed an Android application for hotel self check-in using **Aadhaar-based e-verification** and face verification for seamless guest onboarding.

Research Intern — Centre for Networked Intelligence, IISc Bengaluru Jan 2024 – Jun 2024

- Studied **NVIDIA GPU architecture** and parallel computing concepts for high-performance computing workloads.
- Conducted experiments that outperformed Linux CPU governors in energy efficiency by up to **80%** using the proposed scheduling method.
- Collaborated with **IBM Research** on GPU scheduling techniques and matrix multiplication optimization for parallel architectures.
- Implemented Logistic Regression and Neural Network models in **CUDA C++** for training and inference, and compared their performance with the **PyTorch** framework.

Research Intern — IIIT Guwahati May 2022 – Jul 2022

- Worked on **Brain-Computer Interface (BCI)** systems for EEG signal analysis and classification.
- Studied and implemented **MLP classifiers** and Neural Networks for brain signal processing tasks.

Projects

Classification Simulation (Brain Signal Analysis) Tech: Python, Tkinter

- Engineered a TRL-4 prototype to classify EEG brain signals using `MLPClassifier`, enabling patients with motor neuron disease to communicate with their environment.
- Utilized `EMOTIV INSIGHT` hardware for real-time data collection, classifying test data with **90% accuracy**.

Fashion Recommender Tech: Python, Streamlit, TensorFlow

- Created an application to recommend fashion products using a pre-trained `ResNet` model trained on over 1 million images.
- Implemented a K-Nearest Neighbors algorithm for image similarity, achieving **98% prediction accuracy**.

On-Device Whisper AI Assistant Tech: Kotlin, Jetpack Compose, C++, Whisper

- Developed an on-device AI feature integrating OpenAI's Whisper model for offline speech-to-text transcription on Android Interactive Displays.
- Optimized inference pipeline for low-latency performance, contributing to Samsung's AI Assistant project.

Publications

Choudhury, Rajiv et al. (Apr. 2024). "A Brain-Computer Interface Controlled Assistive Device for Persons with Motor Neuron Disease". In: *2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)* 99.18, pp. 2200–2300. URL: https://ieeexplore.ieee.org/abstract/document/10621489/.

Priya, Aditya et al. (June 2024). "Energy-minimizing workload splitting and frequency selection for guaranteed performance over heterogeneous cores". In: *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems* 99.18, pp. 2200–2300. URL: https://dl.acm.org/doi/pdf/10.1145/3632775.3661968.

Skills

Programming Languages	Python, C++, Java, CUDA C++, C#
Deep Learning Frameworks	TensorFlow, PyTorch
Libraries & Tools	NumPy, Pandas, Scikit-learn, Git, Docker
Platforms / Operating Systems	Tizen, Linux, Windows, Android, nvidia-smi